

Table 1. Summary MGSM multilingual-transfer results for Qwen3-4B, Llama-3.2-3B-Instruct, and Phi-4-mini-instruct, macro-averaged equally across five languages. Values are full-sample means $\pm$ ten-group matched-question jackknife standard errors. Higher is better for every column.

Model	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Original	41.6 $\pm$ 2.5	0.00 $\pm$ 0.00	1.76 $\pm$ 0.03	1.47 $\pm$ 0.03
	A-LQR	22.4 $\pm$ 2.3	<u>1.81 $\pm$ 0.02</u>	1.51 $\pm$ 0.04	1.08 $\pm$ 0.02
	H_∞ (ours)	32.2 $\pm$ 2.4	1.44 $\pm$ 0.03	1.66 $\pm$ 0.03	1.25 $\pm$ 0.04
Llama-3.2-3B	Original	33.0 $\pm$ 1.6	0.00 $\pm$ 0.00	1.68 $\pm$ 0.03	1.44 $\pm$ 0.03
	A-LQR	<u>27.6 $\pm$ 2.6</u>	<u>1.50 $\pm$ 0.03</u>	<u>1.57 $\pm$ 0.04</u>	<u>1.17 $\pm$ 0.04</u>
	H_∞ (ours)	12.2 $\pm$ 2.4	1.47 $\pm$ 0.02	1.02 $\pm$ 0.05	0.91 $\pm$ 0.04
Phi-4-mini	Original	45.4 $\pm$ 2.6	0.00 $\pm$ 0.00	1.79 $\pm$ 0.02	1.52 $\pm$ 0.04
	A-LQR	0.0 $\pm$ 0.0	<u>1.47 $\pm$ 0.04</u>	0.01 $\pm$ 0.01	0.02 $\pm$ 0.01
	H_∞ (ours)	35.2 $\pm$ 2.6	1.21 $\pm$ 0.04	1.56 $\pm$ 0.04	1.05 $\pm$ 0.03